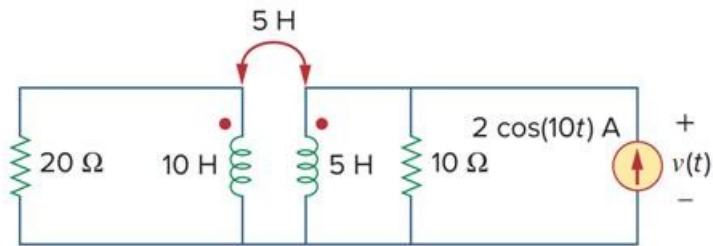


Problem

Find $v(t)$ for the circuit in Fig. 13.77.

Figure 13.77



Step-by-step solution

Step 1 of 11

Refer to Figure 13.77 in the textbook.

Write the standard equation for ac current source.

$$i(t) = I_m \cos \omega t \dots\dots (1)$$

Write the expression for the current source, I .

$$i(t) = 2 \cos 10t \text{ A} \dots\dots (2)$$

Compare equation (1) and (2).

$$\omega = 10 \text{ rad/s}$$

The phasor value of current source is,

$$I = 2 \angle 0^\circ \text{ A}$$

Step 2 of 11

Write the equations for the T-equivalent circuit of the linear transformer.

$$L_a = L_1 - M$$

$$L_b = L_2 - M$$

$$L_c = M$$

Here, L_a , L_b and L_c are the values of inductances of the branches in the T-equivalent circuit.

Calculate the value of L_a .

$$L_a = L_1 - M$$

Substitute 10 H for L_1 and 5 H for M .

$$\begin{aligned} L_a &= 10 - 5 \\ &= 5 \text{ H} \end{aligned}$$

Step 3 of 11

Calculate the value of L_b .

$$L_b = L_2 - M$$

Substitute 5 H for L_2 and 5 H for M .

$$\begin{aligned} L_b &= 5 - 5 \\ &= 0 \text{ H} \end{aligned}$$

Calculate the value of L_c .

$$\begin{aligned} L_c &= M \\ &= 5 \text{ H} \end{aligned}$$

Step 4 of 11

Calculate the impedance of the coil, L_a .

$$Z_a = j\omega L_a$$

Substitute 5 H for L_a and 10 rad/s for ω .

$$\begin{aligned} Z_a &= j\omega L_a \\ &= j(10)(5) \\ &= j50 \Omega \end{aligned}$$

Calculate the impedance of the coil, L_b .

$$Z_b = j\omega L_b$$

Substitute 0 H for L_b and 10 rad/s for ω .

$$\begin{aligned} Z_b &= j(10)(0) \\ &= 0 \Omega \quad (\text{Short circuit}) \end{aligned}$$

Step 5 of 11

Calculate the impedance of the coil, L_c .

$$Z_c = j\omega L_c$$

Substitute 5 H for L_c and 10 rad/s for ω .

$$\begin{aligned} Z_c &= j(10)(5) \\ &= j50 \Omega \end{aligned}$$

Draw the modified circuit in phasor domain.

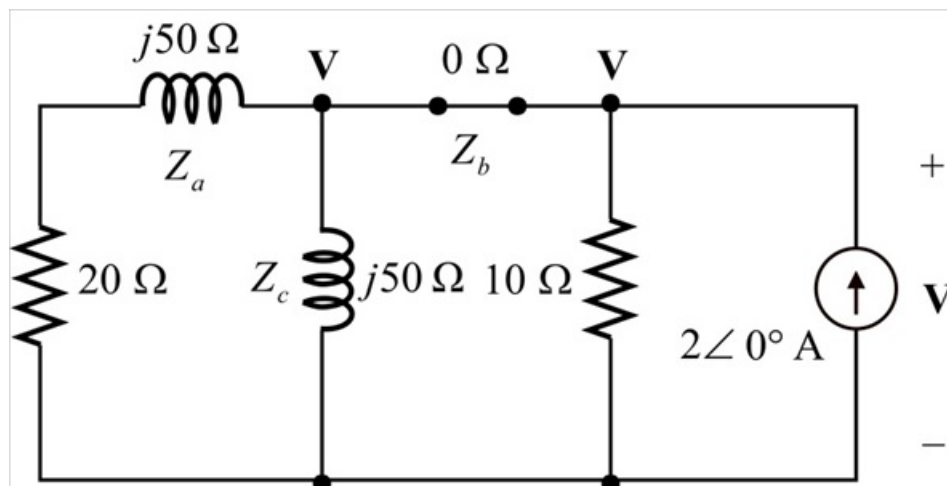


Figure 1

Step 6 of 11

Apply nodal analysis at the node $\mathbf{V}$.

$$\frac{\mathbf{V}}{10} + \frac{\mathbf{V}}{j50} + \frac{\mathbf{V}}{20 + j50} = 2\angle 0^\circ$$

$$\mathbf{V} \left(\frac{1}{10} + \frac{1}{j50} + \frac{1}{20 + j50} \right) = 2$$

$$\mathbf{V}(0.107 - j0.037) = 2$$

$$\mathbf{V} = \frac{2}{0.107 - j0.037}$$

$$\mathbf{V} = 17.668\angle 19.2^\circ \text{ V}$$

Step 7 of 11

Write the time domain expression for $\mathbf{V}$.

$$v(t) = V_m \cos(\omega t + \phi)$$

$$= 17.668 \cos(10t + 19.2^\circ) \text{ V}$$

Therefore, the voltage, $v(t)$ is $17.668 \cos(10t + 19.2^\circ) \text{ V}$.

Step 8 of 11**HTTP Status 404 - Not Found**

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Apache Tomcat/6.0.44**Step 9** of 11**HTTP Status 404 - Not Found**

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Apache Tomcat/6.0.44**Step 10** of 11**HTTP Status 404 - Not Found**

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